

ME780 Assignment 2 Report

by

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Abstract

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I. Part 1: Vehicle Specification

The vehicle selected in CarSim is the generic E-Class SUV. According to the CarSim vehicle assembly dataset, this vehicle has an SLA front suspension and a Hotchkiss solid axle rear suspension. The powertrain is a four-wheel-drive system with a 200 kW internal-combustion engine, a 6-speed automatic transmission, a 1:1 transfer case, and 4.1:1 front and rear differential ratios. The vehicle uses 265/75 R16 tires.

A. Engine Power and Horsepower Conversion

The CarSim powertrain page gives the engine power as $P_{\max} = 200$ kW. Using the conversion $1 \text{ kW} = 1.341 \text{ hp}$, the engine horsepower is $P_{\max} = 200 \times 1.341 = 268.2 \text{ hp}$. Therefore, the vehicle has approximately $P_{\max} = 200 \text{ kW} \approx 268 \text{ hp}$. The maximum engine speed from the engine torque map is $N_{e,\max} = 6700 \text{ rpm}$. Converting this to rad/s,

$$\omega_{e,\max} = N_{e,\max} \frac{2\pi}{60} = 6700 \frac{2\pi}{60} = 701.6 \text{ rad/s}.$$

Thus,

$$\omega_{e,\max} = 6700 \text{ rpm} = 701.6 \text{ rad/s}.$$

The engine page also gives an idle speed of 725 rpm and a minimum speed with clutch of 500 rpm. From the full-throttle torque map, the peak engine torque is approximately

$$T_{e,\max} \approx 413.2 \text{ N} \cdot \text{m}.$$

B. Transmission and Total Gear Ratios

The vehicle uses a 6-speed automatic transmission. The forward gear ratios obtained from CarSim are

$$\beta_i = [3.538, 2.060, 1.404, 1.000, 0.713, 0.582].$$

The final drive ratio is $\beta_d = 4.1$. Therefore, the total gear ratio for each gear is calculated as $\beta_t = \beta_i \beta_d$. For the first gear,

$$\beta_{t,1} = 3.538 \times 4.1 = 14.506.$$

Similarly,

$$\beta_{t,1} = 3.538(4.1) = 14.506,$$

$$\beta_{t,2} = 2.060(4.1) = 8.446,$$

$$\beta_{t,3} = 1.404(4.1) = 5.756,$$

$$\beta_{t,4} = 1.000(4.1) = 4.100,$$

$$\beta_{t,5} = 0.713(4.1) = 2.923,$$

$$\beta_{t,6} = 0.582(4.1) = 2.386.$$

C. Tire Dimensions

The tire dataset used by the vehicle is 265/75 R16. This notation means that the tire section width is 265 mm, the aspect ratio is 75%, and the rim diameter is 16 inches. Therefore, the sidewall height is

$$h_{\text{sidewall}} = 0.75(265) = 198.75 \text{ mm.}$$

The rim diameter is

$$D_{\text{rim}} = 16(25.4) = 406.4 \text{ mm.}$$

Hence, the approximate tire outside diameter is

$$D_{\text{tire}} = D_{\text{rim}} + 2h_{\text{sidewall}} = 406.4 + 2(198.75) = 803.9 \text{ mm.}$$

The corresponding unloaded tire radius is

$$R_{\text{free}} = \frac{803.9}{2} = 401.95 \text{ mm} \approx 402 \text{ mm.}$$

This agrees with the CarSim tire page, which gives the unloaded radius as 402 mm. For performance calculations, the effective rolling radius from CarSim is used:

$$R = 393 \text{ mm} = 0.393 \text{ m.}$$

D. Vehicle Mass and Curb Weight

The sprung mass from the CarSim sprung-mass page is $m_s = 1590 \text{ kg}$. The front unsprung mass is $m_{u,f} = 102.2 \text{ kg}$, and the rear unsprung mass is $m_{u,r} = 115 \text{ kg}$. The tire mass is 28 kg per tire. Therefore, for four tires, $m_{\text{tires}} = 4(28) = 112 \text{ kg}$. The approximate curb mass is calculated by summing the sprung mass, front unsprung mass, rear unsprung mass, and tire masses:

$$m_{\text{curb}} = m_s + m_{u,f} + m_{u,r} + m_{\text{tires}}.$$

Substituting the values,

$$m_{\text{curb}} = 1590 + 102.2 + 115 + 112 = 1919.2 \text{ kg.}$$

Therefore,

$$m_{\text{curb}} \approx 1919.2 \text{ kg.}$$

The corresponding curb weight is $W = m_{\text{curb}}g$. Using $g = 9.81 \text{ m/s}^2$,

$$W = 1919.2(9.81) = 18827.4 \text{ N.}$$

Thus,

$$W \approx 18.83 \text{ kN.}$$

In pounds,

$$m_{\text{curb}} = 1919.2(2.20462) = 4231 \text{ lb.}$$

E. Vehicle Geometry

The CarSim sprung-mass page gives the vehicle target dimensions as

$$\text{Length} = 5500 \text{ mm} = 5.500 \text{ m,}$$

$$\text{Width} = 2176 \text{ mm} = 2.176 \text{ m,}$$

$$\text{Height} = 1714 \text{ mm} = 1.714 \text{ m.}$$

The wheelbase is

$$L = 2950 \text{ mm} = 2.950 \text{ m.}$$

The front suspension and rear suspension kinematics pages give the front and rear track widths as

$$T_f = 1575 \text{ mm} = 1.575 \text{ m,}$$

$$T_r = 1575 \text{ mm} = 1.575 \text{ m.}$$

The displayed ground clearance values are 390 mm at the front and 380 mm at the rear. Therefore, the minimum ground clearance is

$$h_{\text{clearance}} = 380 \text{ mm} = 0.380 \text{ m.}$$

The sprung-mass CG height is

$$h_{\text{CG}} = 720 \text{ mm} = 0.720 \text{ m.}$$

The longitudinal distance from the front axle to the CG is

$$a = 1180 \text{ mm} = 1.180 \text{ m.}$$

Since the wheelbase is $L = 2.950 \text{ m}$, the distance from the CG to the rear axle is

$$b = L - a = 2.950 - 1.180 = 1.770 \text{ m.}$$

Thus,

$$a = 1.180 \text{ m,} \quad b = 1.770 \text{ m.}$$

F. Aerodynamic Parameters

From the CarSim aerodynamic dataset, the aerodynamic drag coefficient at zero aerodynamic slip angle is $C_d = 0.30$. The frontal area is $A_f = 3.0 \text{ m}^2$. The air density used in the CarSim aerodynamic model

is $\rho = 1.206 \text{ kg/m}^3$. The aerodynamic reference length is equal to the wheelbase:

$$L_{\text{ref}} = 2950 \text{ mm} = 2.950 \text{ m}.$$

G. Summary of Extracted Vehicle Specifications

Table 1: CarSim E-Class SUV specifications extracted for Assignment 2.

Parameter	Value
Vehicle model	Generic E-Class SUV
Suspension type	SLA front suspension, Hotchkiss solid axle rear suspension
Drive type	Four-wheel drive
Engine power	200 kW
Horsepower	268.2 hp
Maximum engine speed	6700 rpm = 701.6 rad/s
Idle speed	725 rpm
Minimum speed with clutch	500 rpm
Peak engine torque from visible map	$\approx 413.2 \text{ N} \cdot \text{m}$
Transmission	6-speed automatic
Reverse gear ratio	-3.168
1st gear ratio	3.538
2nd gear ratio	2.060
3rd gear ratio	1.404
4th gear ratio	1.000
5th gear ratio	0.713
6th gear ratio	0.582
Transfer case ratio	1:1
Final drive ratio	4.1:1
Total ratio, 1st gear	14.506
Total ratio, 2nd gear	8.446
Total ratio, 3rd gear	5.756
Total ratio, 4th gear	4.100
Total ratio, 5th gear	2.923
Total ratio, 6th gear	2.386
Tire size	265/75 R16
Tire width	265 mm
Effective rolling radius	393 mm = 0.393 m
Unloaded/free tire radius	402 mm = 0.402 m

Parameter	Value
Tire mass	28 kg per tire
Tire vertical spring rate	502 N/mm
Reference vertical force	11500 N
Contact patch length	85 mm
Contact patch width	265 mm
Wheelbase	2950 mm = 2.950 m
Overall length	5500 mm = 5.500 m
Overall width	2176 mm = 2.176 m
Overall height	1714 mm = 1.714 m
Front track width	1575 mm = 1.575 m
Rear track width	1575 mm = 1.575 m
Front ground clearance	390 mm
Rear ground clearance	380 mm
Minimum ground clearance	380 mm = 0.380 m
Sprung mass	1590 kg
Front unsprung mass	102.2 kg
Rear unsprung mass	115 kg
Total tire mass	112 kg
Estimated curb mass	1919.2 kg
Estimated curb weight	18.83 kN
Estimated curb weight in lb	4231 lb
CG height	720 mm = 0.720 m
Distance from front axle to CG, a	1180 mm = 1.180 m
Distance from CG to rear axle, b	1770 mm = 1.770 m
Roll inertia, I_{xx}	894.4 kg · m ²
Pitch inertia, I_{yy}	2687.1 kg · m ²
Yaw inertia, I_{zz}	2687.1 kg · m ²
Drag coefficient at zero aero slip angle	$C_d = 0.30$
Frontal area	$A_f = 3.0 \text{ m}^2$
Air density used in CarSim	$\rho = 1.206 \text{ kg/m}^3$
Aerodynamic reference length	2950 mm = 2.950 m

II. Part 2: Performance Metrics Prediction

A. Vehicle Parameters Used

The vehicle parameters used for the performance calculations were extracted from the CarSim E-Class SUV dataset in Part 1. The vehicle mass, rolling radius, aerodynamic parameters, rolling resistance co-

efficient, and drivetrain ratios are summarized below: $m = 1919.2$ kg, $R = 0.393$ m, $C_d = 0.30$, $A_f = 3.0$ m², $\rho = 1.206$ kg/m³, $f_r = 0.0044$, $g = 9.81$ m/s². The transmission gear ratios are

$$\beta_i = [3.538, 2.060, 1.404, 1.000, 0.713, 0.582].$$

The final drive ratio is $\beta_d = 4.1$. Since the given assumption is 100% efficiency for the transmission and differential, the total gear ratio in each gear is $\beta_t = \beta_i \beta_d$. Therefore,

$$\beta_{t,1} = 3.538(4.1) = 14.506,$$

$$\beta_{t,2} = 2.060(4.1) = 8.446,$$

$$\beta_{t,3} = 1.404(4.1) = 5.756,$$

$$\beta_{t,4} = 1.000(4.1) = 4.100,$$

$$\beta_{t,5} = 0.713(4.1) = 2.923,$$

$$\beta_{t,6} = 0.582(4.1) = 2.386.$$

The maximum engine speed from the CarSim engine map is $N_{e,\max} = 6700$ rpm. Converting this value to rad/s, $\omega_{e,\max} = N_{e,\max} \frac{2\pi}{60} = 6700 \frac{2\pi}{60} = 701.6$ rad/s.

B. Parabolic Approximation of Engine Torque Curve

The full-throttle engine torque data were taken from the CarSim engine torque table using the column corresponding to throttle position equal to 1. The engine torque characteristic was approximated using the parabolic form

$$T_e(\omega_e) = A_0 + A_1\omega_e + A_2\omega_e^2,$$

where T_e is the engine torque in N · m, and ω_e is the engine speed in rad/s. The positive full-throttle operating region from idle speed to 6100 rpm was used for the curve fitting. The fitted parabolic torque curve is

$$T_e(\omega_e) = 131.672 + 1.46058\omega_e - 1.88563 \times 10^{-3}\omega_e^2$$

where ω_e is in rad/s. Equivalently, using engine speed in rpm,

$$T_e(N_e) = 131.672 + 0.152952N_e - 2.06782 \times 10^{-5}N_e^2$$

where N_e is in rpm. The maximum torque from the fitted curve is $T_{e,\max} = 414.51$ N · m. This occurs at approximately $N_e = 3698$ rpm. The corresponding fitted peak engine power is approximately $P_{\max} = 200.63$ kW. This is consistent with the nominal 200 kW engine in CarSim.

C. Maximum Speed Calculation

The engine speed and vehicle speed are related by $\omega_e = \frac{\beta_t}{R}u$, where u is the vehicle speed in m/s. The tractive force at the wheels is $F_t = \frac{\beta_t}{R}T_e(\omega_e)$. Substituting the relation between engine speed and vehicle

speed gives

$$F_t(u) = \frac{\beta_t}{R} T_e \left(\frac{\beta_t}{R} u \right).$$

The resistance force on a flat road is

$$F_R = f_r mg + \frac{1}{2} \rho C_d A_f u^2.$$

Substituting the numerical values,

$$F_R = 0.0044(1919.2)(9.81) + \frac{1}{2}(1.206)(0.30)(3.0)u^2.$$

Therefore,

$$F_R = 82.84 + 0.5427u^2$$

where u is in m/s and F_R is in N. At maximum speed, the vehicle acceleration is zero: $\dot{u} = 0$. Therefore, $F_t = F_R$. For each gear, the equation $F_t(u) - F_R(u) = 0$ was solved, while also satisfying the engine speed limit $\omega_e \leq \omega_{e,\max}$. The maximum speed results for each gear are shown in Table 2.

Table 2: Maximum speed calculation for each gear.

Gear	Total ratio β_t	Limiting condition	Maximum speed
1	14.506	Redline limited	68.43 km/h
2	8.446	Redline limited	117.53 km/h
3	5.756	Redline limited	172.44 km/h
4	4.100	Force limited	239.29 km/h
5	2.923	Force limited	255.14 km/h
6	2.386	Force limited	240.89 km/h

The highest maximum speed occurs in 5th gear. Therefore, the predicted maximum speed is $u_{\max} = 255.14$ km/h or equivalently, $u_{\max} = 70.87$ m/s.

D. Acceleration Time from 0 to 100 km/h

The longitudinal equation of motion is $F_t - F_R = m_d \dot{u}$, where m_d is the effective dynamic mass including rotating inertias. The effective dynamic mass in each gear is approximated as

$$m_{d,i} = m + \frac{I_e \beta_{t,i}^2}{R^2} + \frac{4I_w}{R^2}.$$

From the CarSim data, $I_e = 0.3 \text{ kg} \cdot \text{m}^2$ and $I_w = 2.8 \text{ kg} \cdot \text{m}^2$.

The resulting dynamic masses are given in Table 3.

The acceleration is

$$\dot{u} = \frac{F_t - F_R}{m_d}.$$

Table 3: Effective dynamic mass in each gear.

Gear	m_d
1	2400.43 kg
2	2130.28 kg
3	2056.08 kg
4	2024.37 kg
5	2008.31 kg
6	2002.78 kg

Therefore,

$$dt = \frac{m_d}{F_t - F_R} du.$$

The acceleration time from 0 to 100 km/h is

$$t_{0-100} = \int_0^{27.78} \frac{m_d}{F_t(u) - F_R(u)} du,$$

because 100 km/h = 27.78 m/s.

Using maximum-acceleration gear selection, the vehicle starts in 1st gear and shifts to 2nd gear at approximately $u = 64.67$ km/h. The calculated time in 1st gear is $t_1 = 3.73$ s. The calculated time in 2nd gear, from 64.67 km/h to 100 km/h, is $t_2 = 2.65$ s. Therefore,

$$t_{0-100} = t_1 + t_2.$$

$$t_{0-100} = 3.73 + 2.65.$$

$$t_{0-100} = 6.38 \text{ s.}$$

E. Maximum Gradability

For maximum gradability, low-speed operation is considered. Therefore, the aerodynamic drag force is neglected: $F_D \approx 0$. The maximum tractive force occurs in 1st gear:

$$F_{t,\max} = \frac{\beta_{t,1} T_{e,\max}}{R}.$$

Using $\beta_{t,1} = 14.506$, $T_{e,\max} = 414.51 \text{ N} \cdot \text{m}$, and $R = 0.393 \text{ m}$, the maximum tractive force is

$$F_{t,\max} = \frac{14.506(414.51)}{0.393}.$$

$$F_{t,\max} = 15299.68 \text{ N.}$$

The vehicle weight is $W = mg = 1919.2(9.81) = W = 18827.35 \text{ N}$. For climbing a grade, the required longitudinal force is

$$F_{\text{grade}} = mg \sin \theta + f_r mg \cos \theta.$$

The powertrain-limited maximum grade angle is found from

$$F_{t,\max} = mg \sin \theta + f_r mg \cos \theta.$$

Solving this equation gives

$$\theta_{\max, \text{powertrain}} = 54.10^\circ.$$

The corresponding gradeability percentage is

$$\text{Gradeability} = \tan(\theta) \times 100.$$

Therefore,

$$\text{Gradeability}_{\text{powertrain}} = \tan(54.10^\circ) \times 100.$$

$$\text{Gradeability}_{\text{powertrain}} = 138.15\%.$$

However, the maximum gradability must also satisfy the tire-road friction limit. For a four-wheel-drive vehicle, the friction-limited climbing condition is

$$\mu mg \cos \theta = mg \sin \theta + f_r mg \cos \theta.$$

Dividing both sides by $mg \cos \theta$,

$$\mu = \tan \theta + f_r.$$

Therefore,

$$\tan \theta = \mu - f_r.$$

E..1 Dry Road, $\mu = 1$

For dry road, $\mu = 1$. Thus,

$$\tan \theta = 1 - 0.0044 = 0.9956.$$

$$\theta_{\max, \text{dry}} = \tan^{-1}(0.9956).$$

$$\theta_{\max, \text{dry}} = 44.87^\circ.$$

The corresponding gradeability is

$$\text{Gradeability}_{\text{dry}} = 0.9956 \times 100.$$

$$\text{Gradeability}_{\text{dry}} = 99.56\%.$$

Since the dry-road friction-limited value is less than the powertrain-limited value, the final dry-road gradability is

$$\text{Final dry-road gradability} = 99.56\%.$$

E..2 Wet Asphalt, $\mu = 0.5$

For wet asphalt, $\mu = 0.5$. Thus,

$$\tan \theta = 0.5 - 0.0044 = 0.4956.$$

$$\theta_{\max, \text{wet}} = \tan^{-1}(0.4956).$$

$$\theta_{\max, \text{wet}} = 26.36^\circ.$$

The corresponding gradeability is

$$\text{Gradeability}_{\text{wet}} = 0.4956 \times 100.$$

$$\text{Gradeability}_{\text{wet}} = 49.56\%.$$

Since the wet-road friction-limited value is less than the powertrain-limited value, the final wet-asphalt gradability is

$$\text{Final wet-asphalt gradability} = 49.56\%.$$

F. Comparison with CarSim Result

The CarSim full-throttle acceleration plot was used to obtain the 0–100 km/h acceleration time. From the CarSim speed-time plot, the vehicle reaches 100 km/h at

$$t_{0-100, \text{CarSim}} = 6.4475 \text{ s.}$$

The analytical result is

$$t_{0-100, \text{analytical}} = 6.38 \text{ s.}$$

The absolute difference is

$$\Delta t = 6.4475 - 6.38 = 0.0675 \text{ s.}$$

The percentage difference is

$$\% \text{ difference} = \frac{|t_{\text{analytical}} - t_{\text{CarSim}}|}{t_{\text{CarSim}}} \times 100.$$

$$\% \text{ difference} = \frac{|6.38 - 6.4475|}{6.4475} \times 100.$$

$$\% \text{ difference} = 1.05\%.$$

Therefore, the analytical prediction agrees closely with the CarSim result for the 0–100 km/h acceleration time. The small difference is expected because the analytical model uses simplified longitudinal dynamics, a parabolic approximation of the engine torque curve, ideal drivetrain efficiency, and simplified rotating inertia effects. CarSim includes more detailed drivetrain, tire, shifting, and vehicle dynamics

models.

The CarSim plot available for this comparison covers only the first 10 seconds of the acceleration test. Since the vehicle is still accelerating at the end of that plot, a valid CarSim maximum speed could not be extracted from the available result. Therefore, the maximum speed is reported from the analytical calculation.

G. Summary of Part 2 Results

Table 4: Summary of Part 2 performance predictions and CarSim comparison.

Metric	Analytical result	CarSim result
Parabolic torque curve	$T_e(\omega_e) = 131.672 + -$ $1.46058\omega_e - 1.88563 \times$ $10^{-3}\omega_e^2$	
Maximum speed	255.14 km/h	Not available from 10 s plot
0–100 km/h acceleration time	6.38 s	6.4475 s
Difference in 0–100 km/h time	1.05%	–
Powertrain-limited gradability	$54.10^\circ = 138.15\%$	Not simulated
Dry-road final gradability, $\mu = 1$	$44.87^\circ = 99.56\%$	Not simulated
Wet-asphalt final gradability, $\mu = 0.5$	$26.36^\circ = 49.56\%$	Not simulated